

Transformations of Quadratics in Vertex Form

Comprehensive Notes and Practice

1 Core Concept: Vertex Form

The **vertex form** of a quadratic function is:

$$y = a(x - h)^2 + k \tag{1}$$

Where:

- (h, k) is the **vertex** of the parabola
- a controls the **direction** and **width** of the parabola

2 Transformation Examples

Worked Examples	
Example 1: $y = 2(x - 3)^2 + 1$	Example 2: $y = -\frac{1}{2}(x + 4)^2 - 2$
Starting from $y = x^2$:	Starting from $y = x^2$:
1. Vertical stretch by 2 ($a = 2$) → Narrower parabola	1. Vertical compression by $\frac{1}{2}$ ($ a = \frac{1}{2}$) → Wider parabola
2. Shift right 3 units ($h = 3$) → Vertex: $(0, 0) \rightarrow (3, 0)$	2. Reflection over x-axis ($a < 0$) → Opens downward
3. Shift up 1 unit ($k = 1$) → Vertex: $(3, 0) \rightarrow (3, 1)$	3. Shift left 4 units ($h = -4$) → Vertex: $(0, 0) \rightarrow (-4, 0)$
	4. Shift down 2 units ($k = -2$) → Vertex: $(-4, 0) \rightarrow (-4, -2)$
Final vertex: $(3, 1)$ Opens: Upward ($a > 0$)	Final vertex: $(-4, -2)$ Opens: Downward ($a < 0$)

Transformation	Effect	Example
$a > 1$	Narrower (vertical stretch)	$y = 3x^2$
$0 < a < 1$	Wider (vertical compression)	$y = 0.25x^2$
$a < 0$	Opens down (reflection)	$y = -x^2$
$(x - h), h > 0$	Shift right h units	$y = (x - 5)^2$
$(x + h), h > 0$	Shift left h units	$y = (x + 5)^2$
$+k$	Shift up k units	$y = x^2 + 7$
$-k$	Shift down k units	$y = x^2 - 3$

3 Practice Problems

3.1 Section A: Identifying Transformations

For each function, identify the vertex, direction (up/down), and describe all transformations from $y = x^2$.

1. $y = 3(x - 2)^2 + 5$

Answer:

Vertex: _____ Opens: _____

2. $y = -(x + 1)^2 - 4$

Answer:

Vertex: _____ Opens: _____

3. $y = \frac{1}{2}(x - 6)^2$

Answer:

Vertex: _____ Opens: _____

4. $y = -2(x + 3)^2 + 1$

Answer:

Vertex: _____ Opens: _____

5. $y = (x + 5)^2 - 7$

Answer:

Vertex: _____ Opens: _____

6. $y = 4(x - 1)^2 - 2$

Answer:

Vertex: _____ Opens: _____

7. $y = -\frac{1}{4}(x + 2)^2 + 3$

Answer:

Vertex: _____ Opens: _____

8. $y = -(x - 4)^2$

Answer:

Vertex: _____ Opens: _____

3.2 Section B: Writing Equations from Descriptions

Write the vertex form equation for each description.

9. Vertex at
- $(2, 3)$
- , opens upward, vertical stretch by factor of 2

Answer:

Answer: _____

10. Vertex at
- $(-1, -5)$
- , opens downward, no vertical stretch/compression

Answer:

Answer: _____

11. Vertex at $(0, 4)$, opens upward, vertical compression by factor of $\frac{1}{3}$

Answer:

Answer: _____

12. Vertex at $(7, -2)$, opens downward, vertical stretch by factor of 5

Answer:

Answer: _____

13. The parent function $y = x^2$ shifted left 3 units and down 6 units

Answer:

Answer: _____